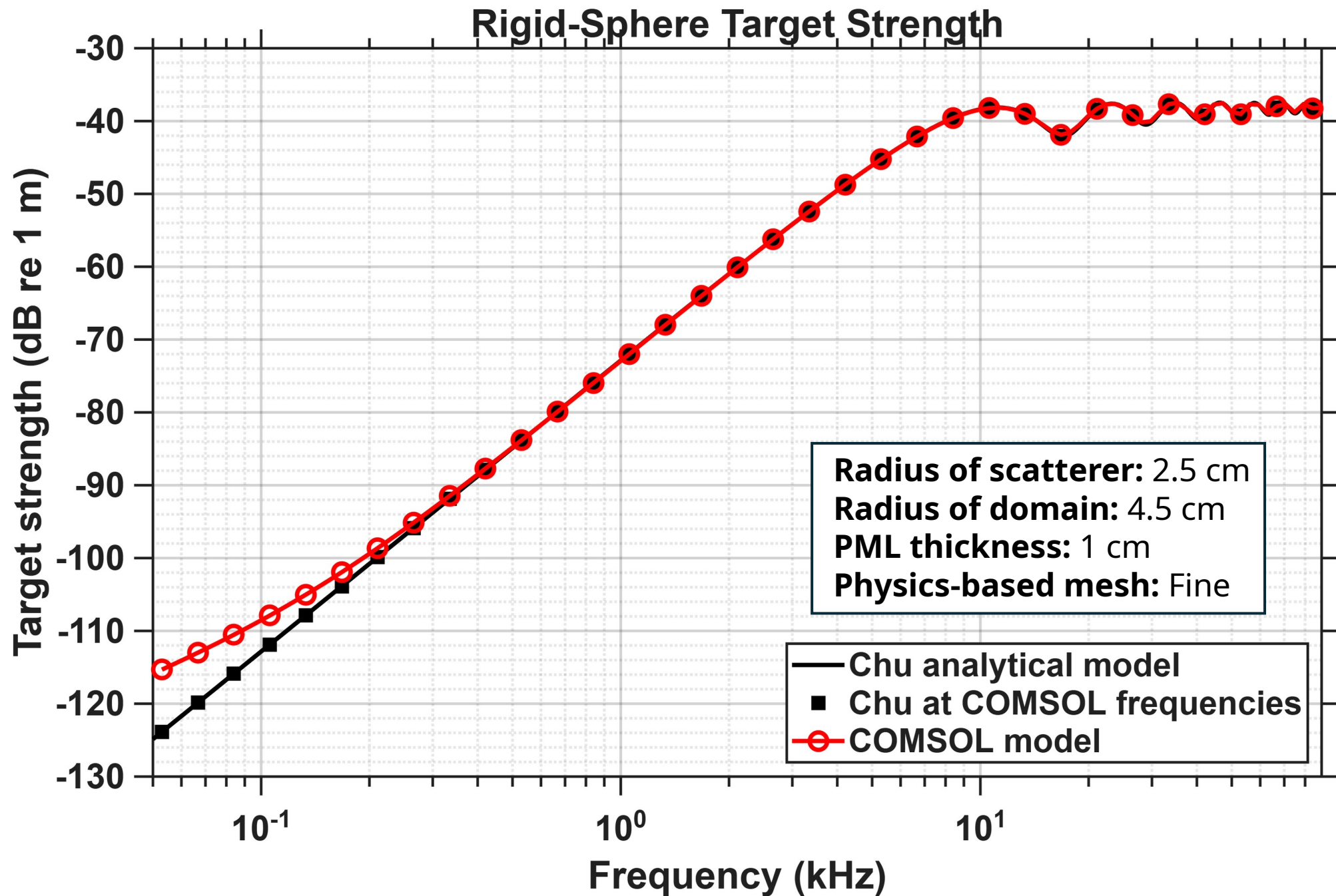
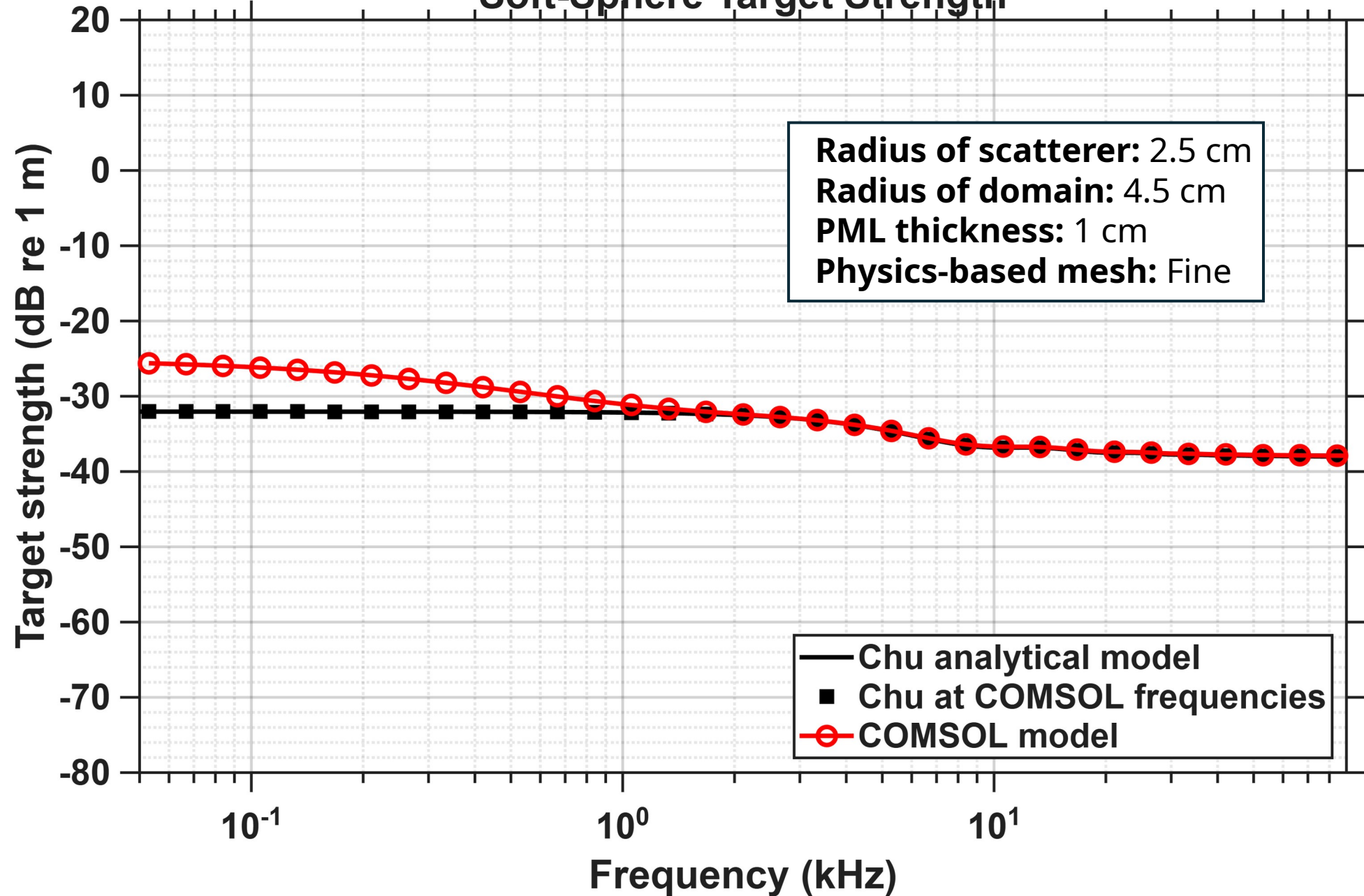


# Acoustic scattering by spheres and swim bladders

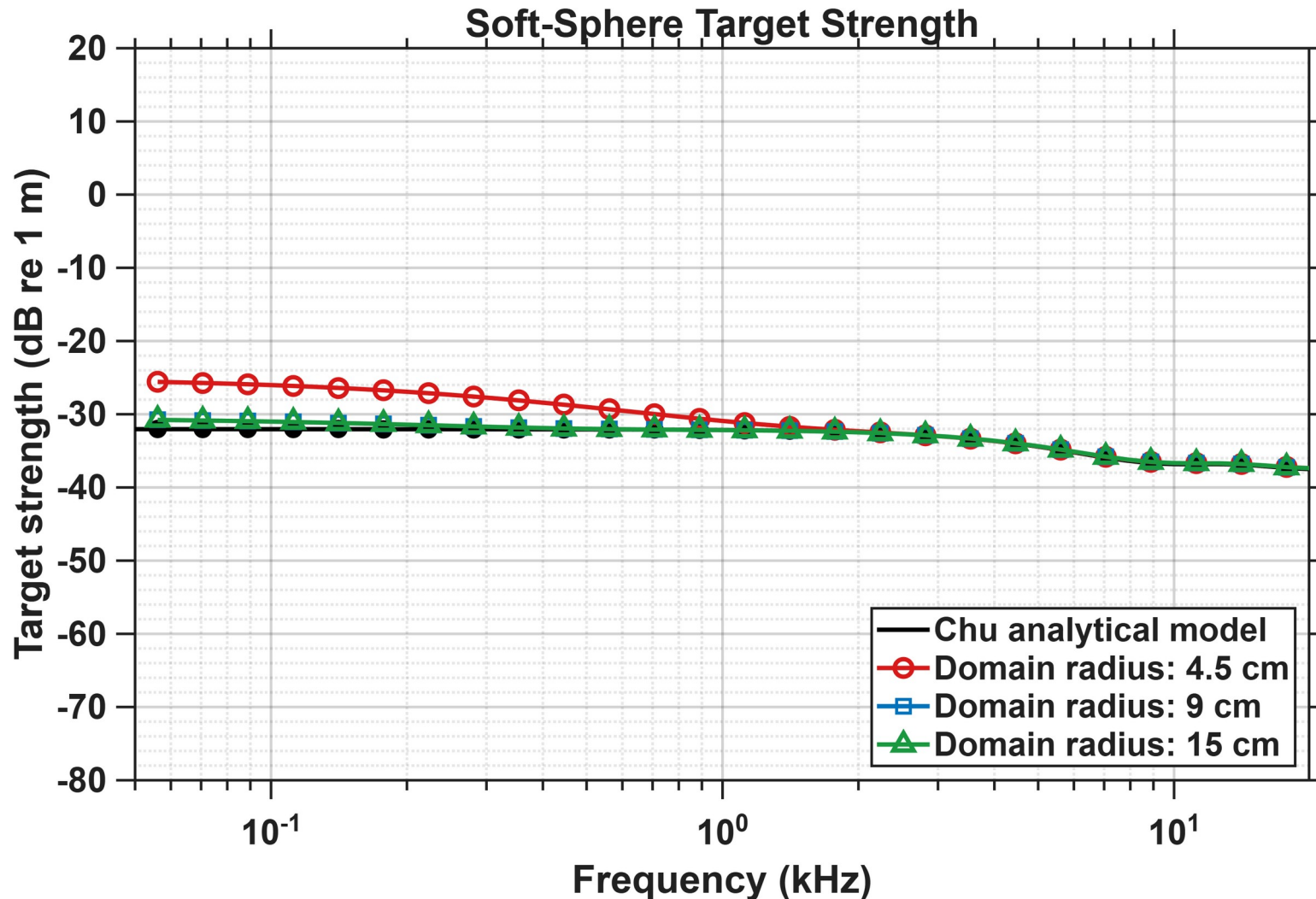
# **Spherical benchmark models**



## Soft-Sphere Target Strength

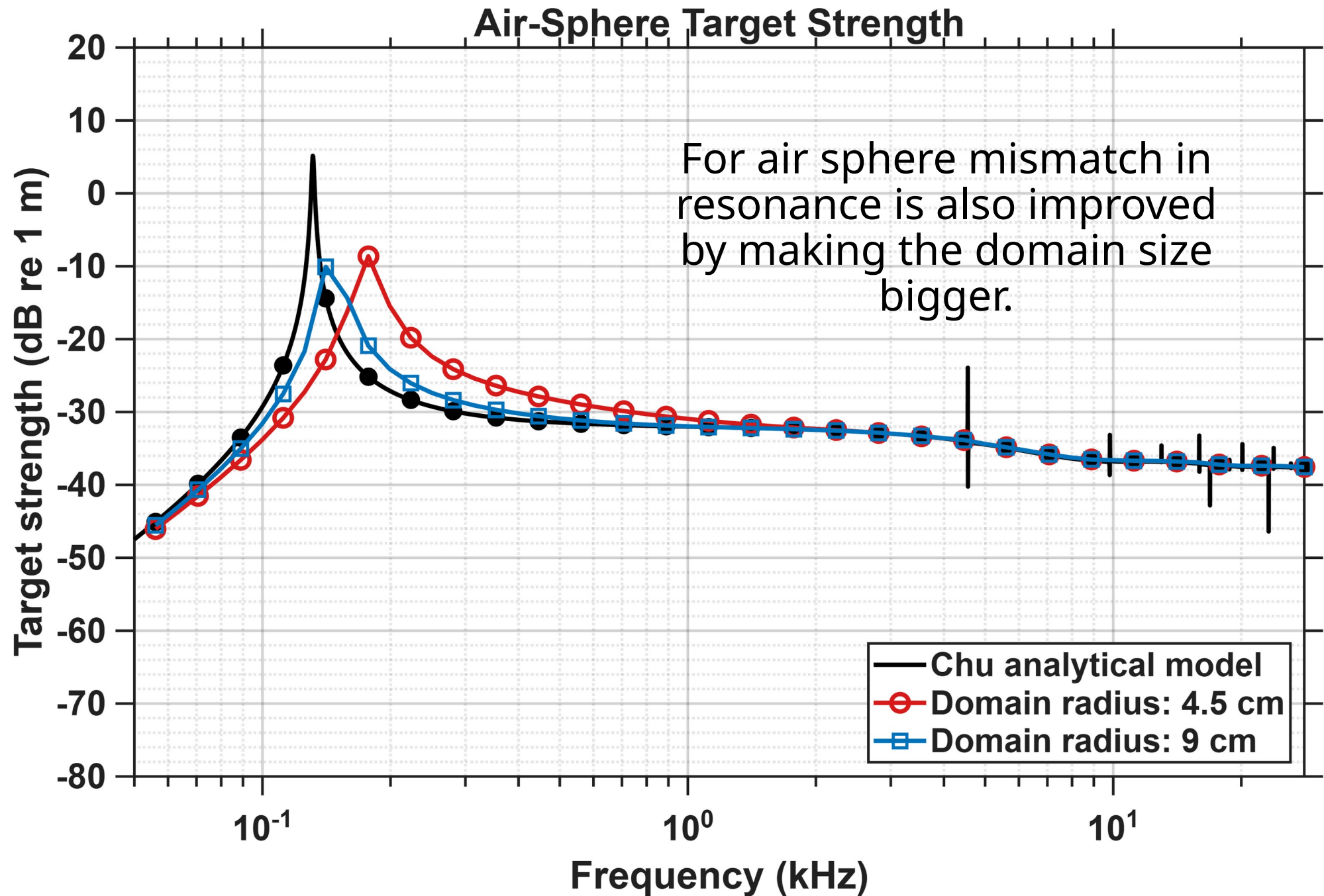


# Low frequency mismatches are 'mostly' resolved by making the domain size bigger



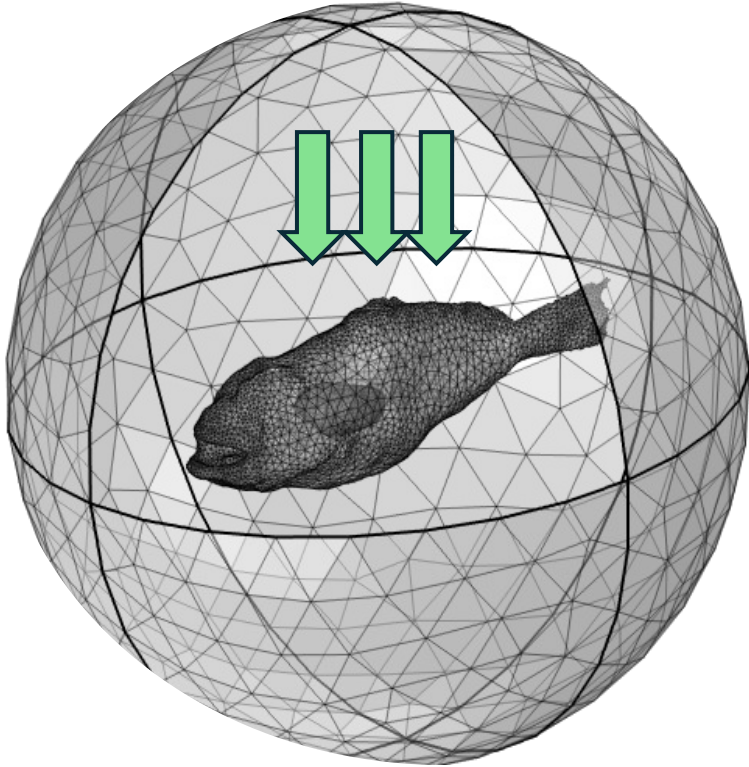
## Note:

- 1) The limits on the y-axis are different compared to the previous soft sphere plot to see differences more clearly
- 2) These simulations only focus on the low frequency region where the mismatches were

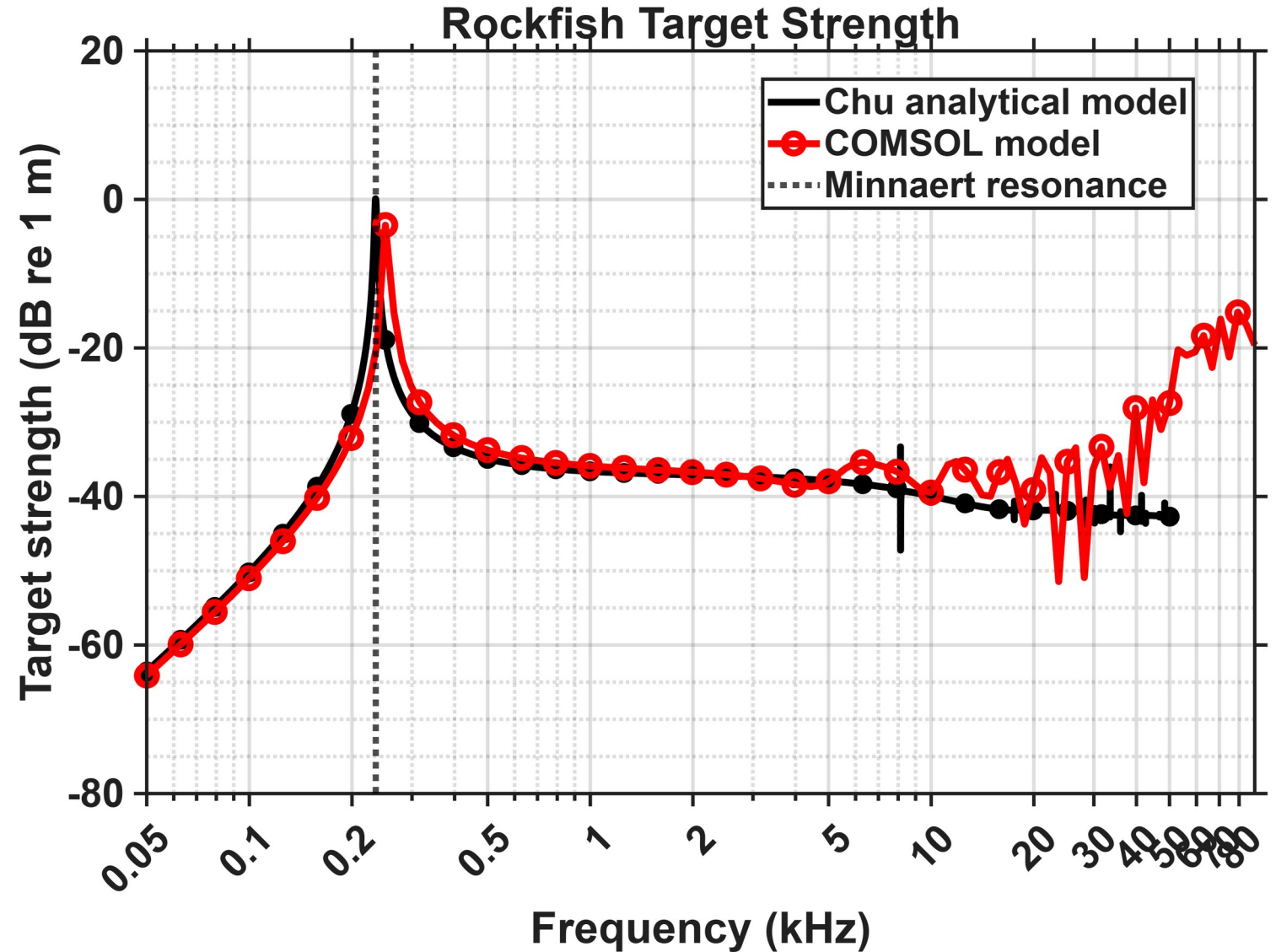


# **Fish and swim bladder models**

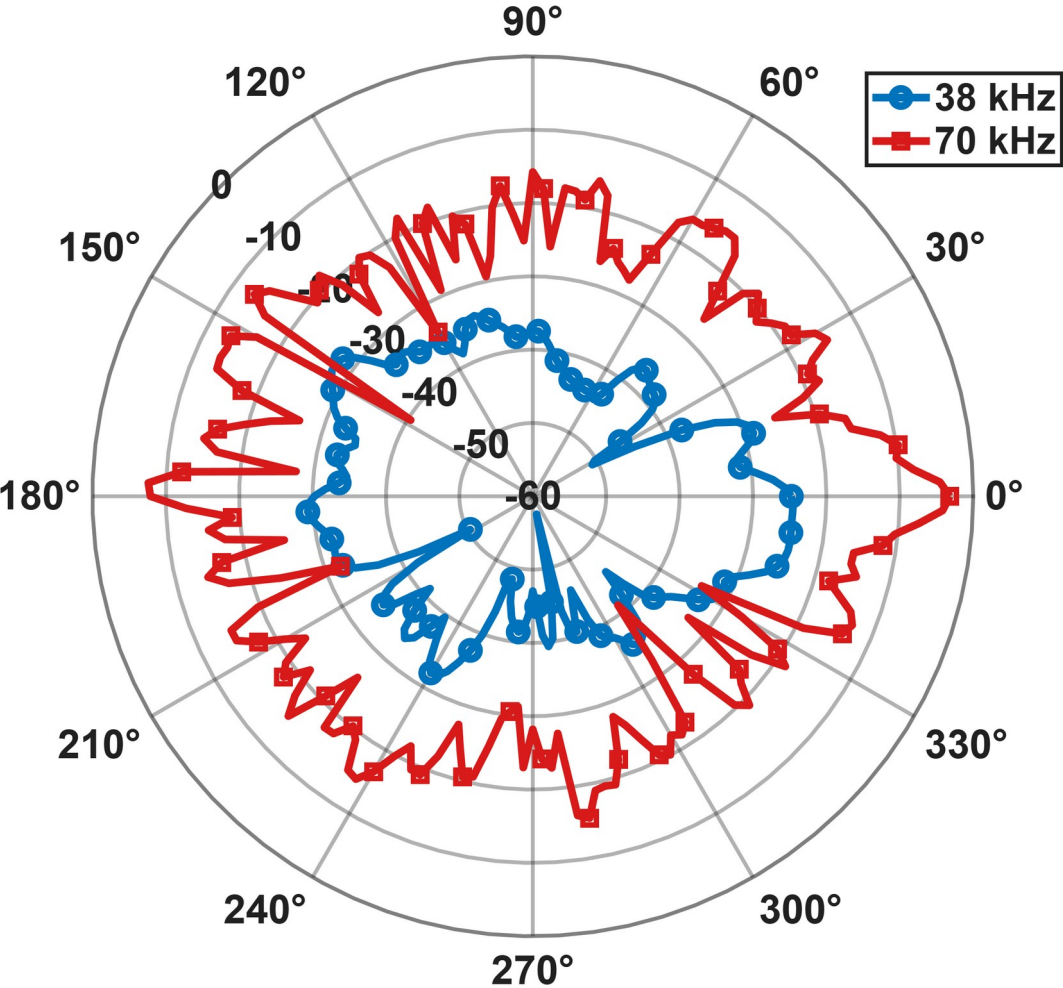
# Rockfish: Dorso-ventral incidence



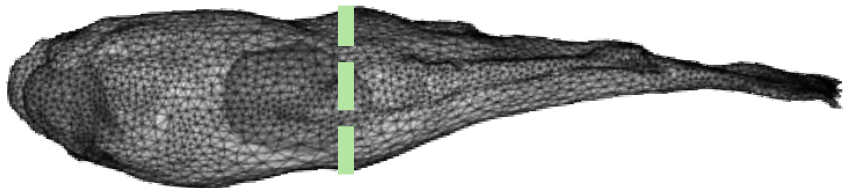
**Water domain radius:** 15 cm  
**PML thickness:** 2.5 cm  
**Volume-equivalent spherical radius of swim bladder:** 1.4 cm



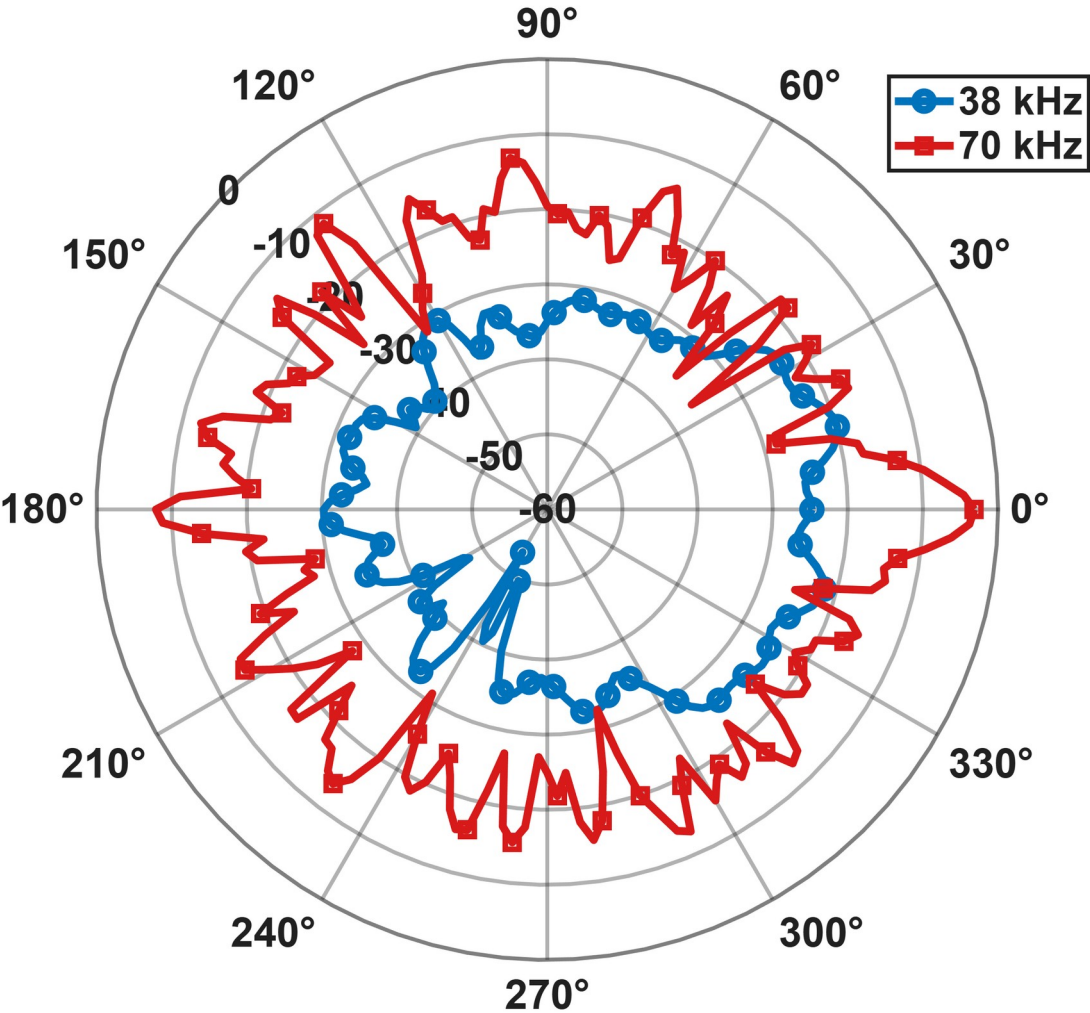
Rockfish Azimuthal Radiation Pattern



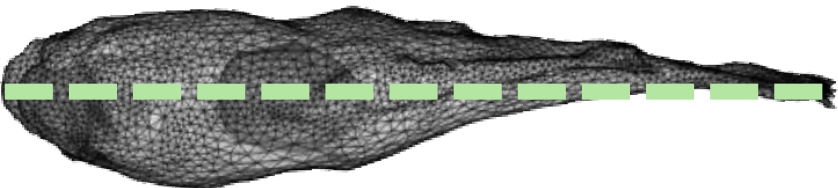
Target strength (dB re 1 m)



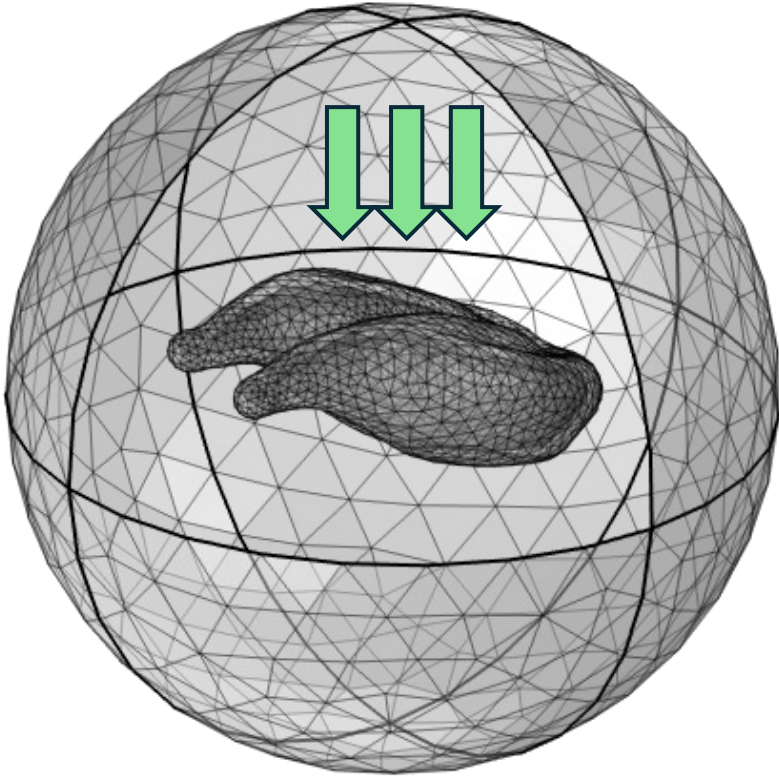
Rockfish Vertical Radiation Pattern



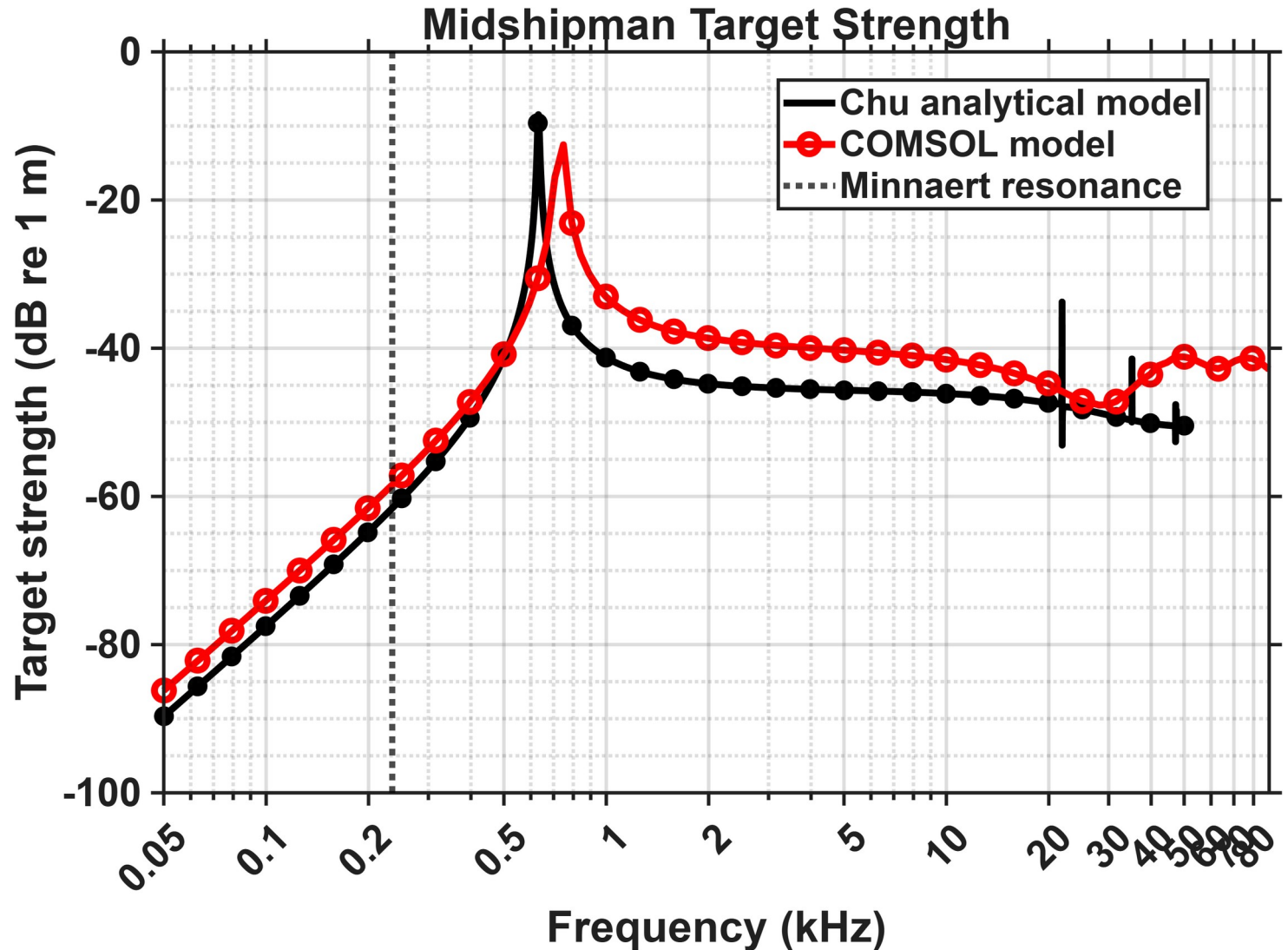
Target strength (dB re 1 m)



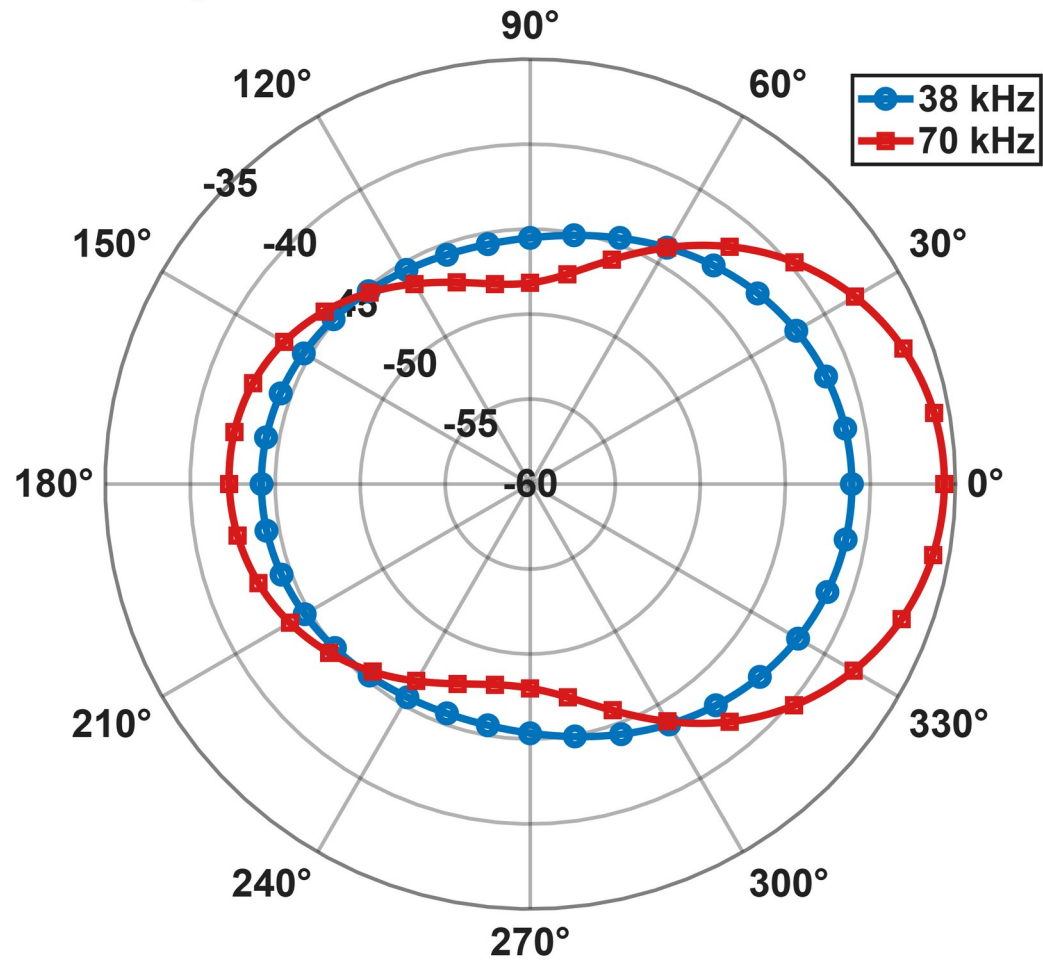
# Midshipman: Dorso-ventral incidence



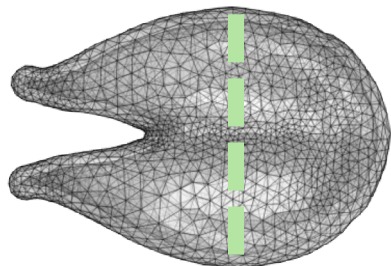
**Water domain radius:** 2 cm  
**PML thickness:** 0.5 cm  
**Volume-equivalent spherical radius of swim bladder:** 0.52 cm



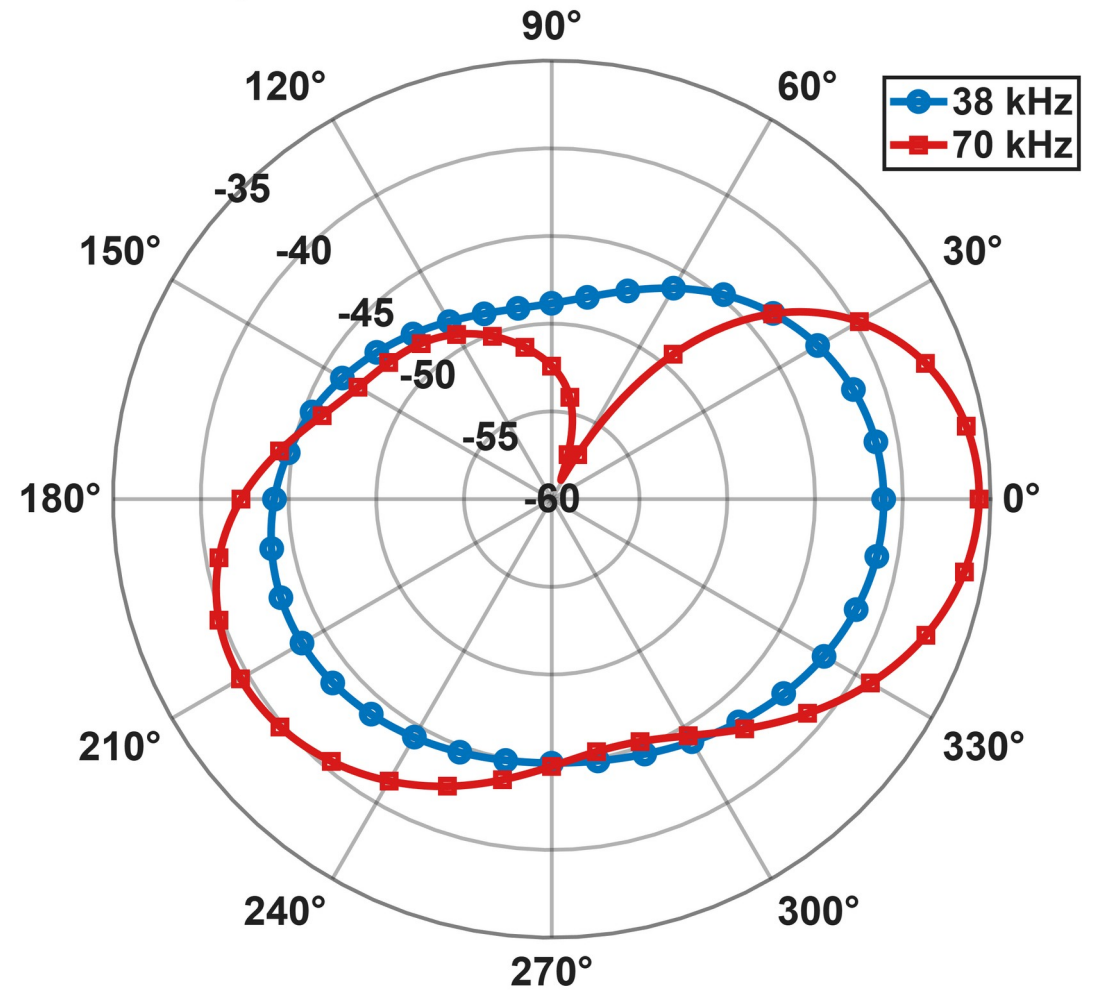
# Midshipman Azimuthal Radiation Pattern



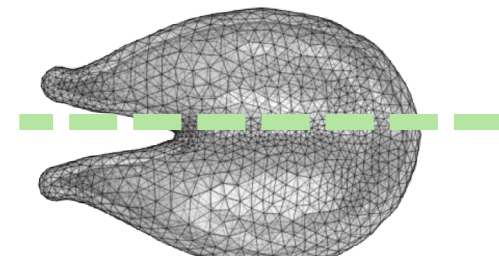
Target strength (dB re 1 m)



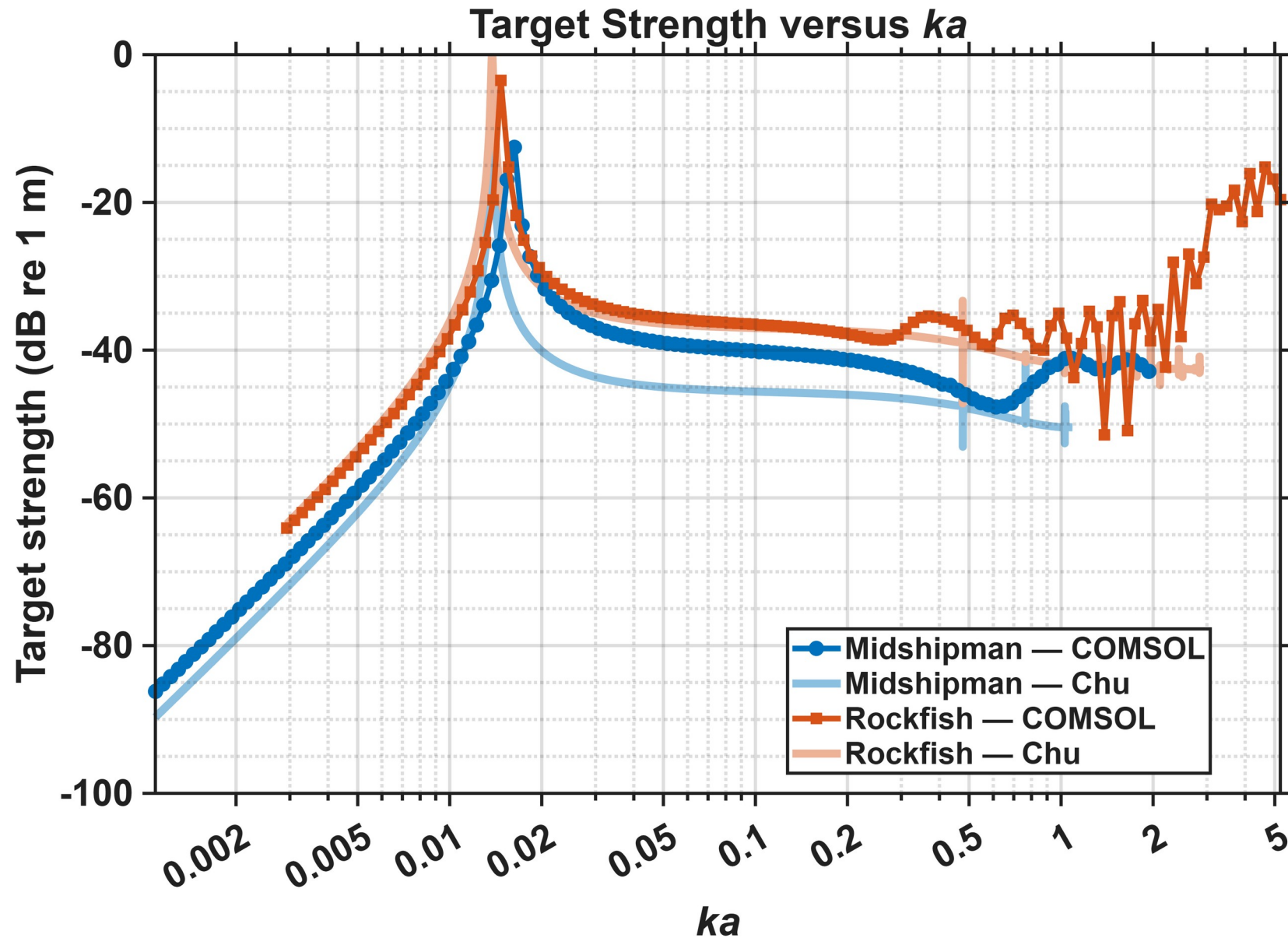
# Midshipman Vertical Radiation Pattern



Target strength (dB re 1 m)



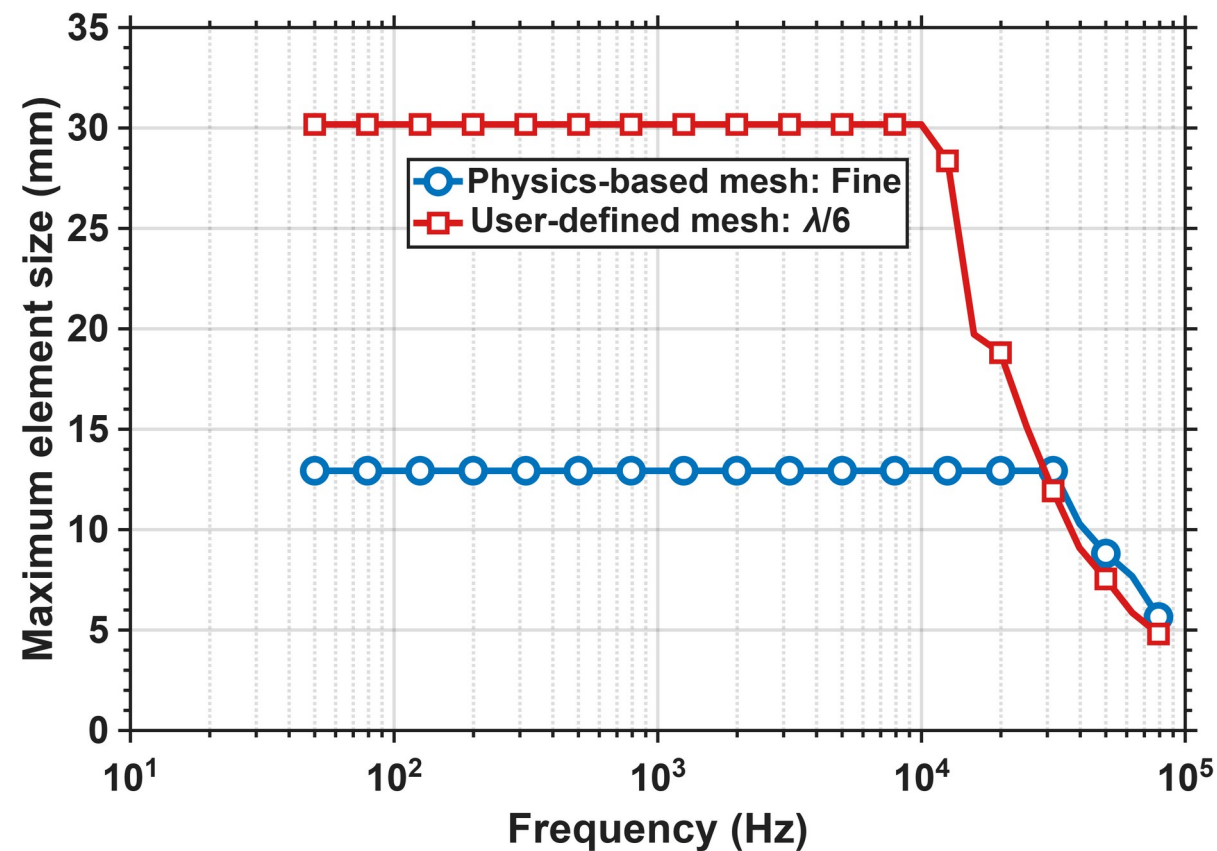
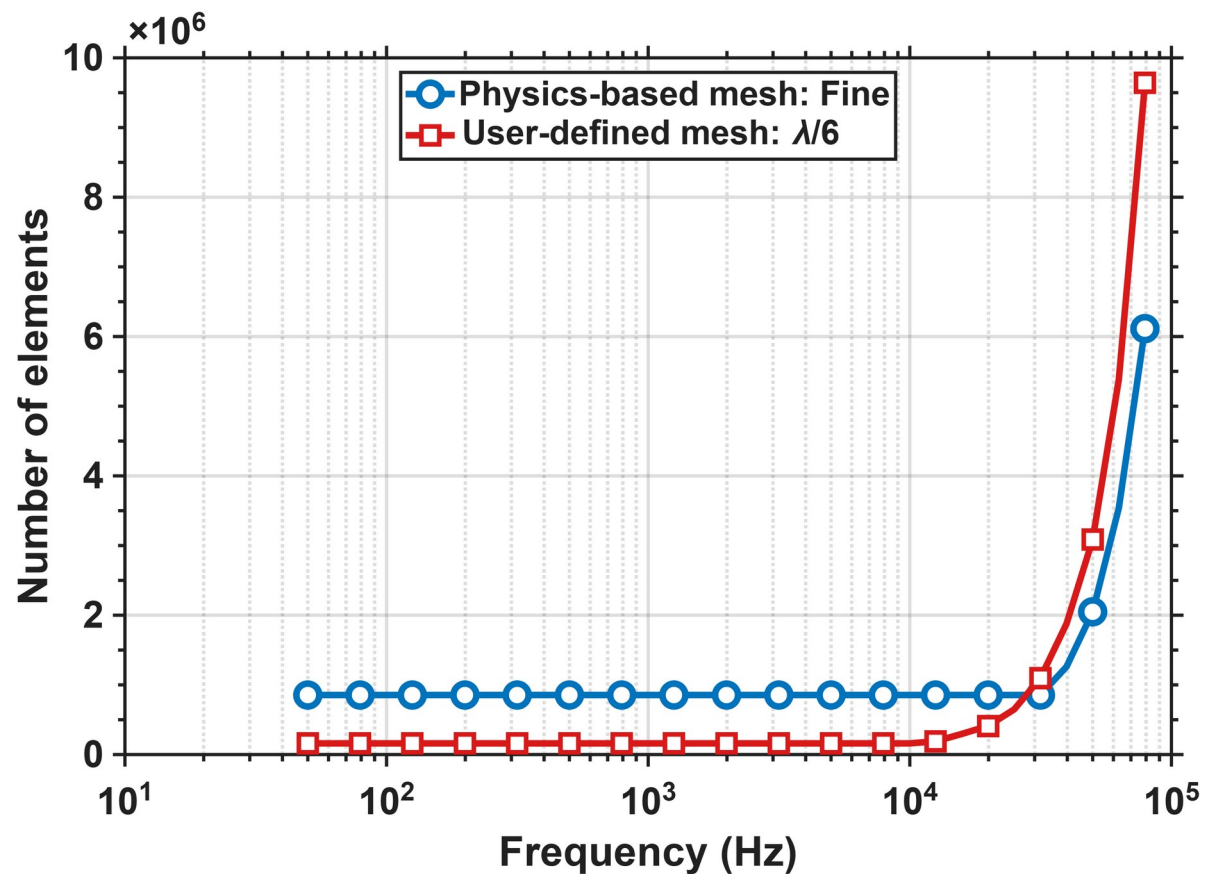
# Midshipman-Rockfish dorsoventral incidence



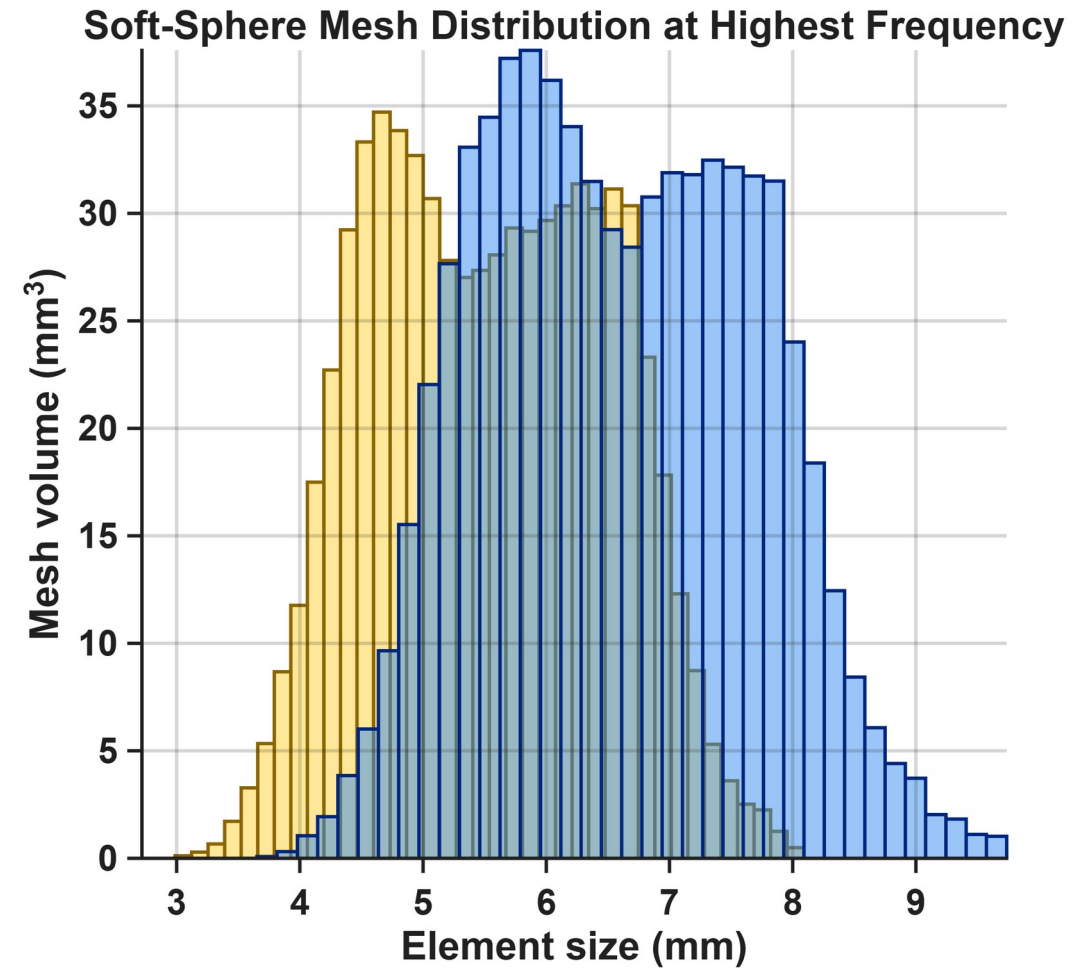
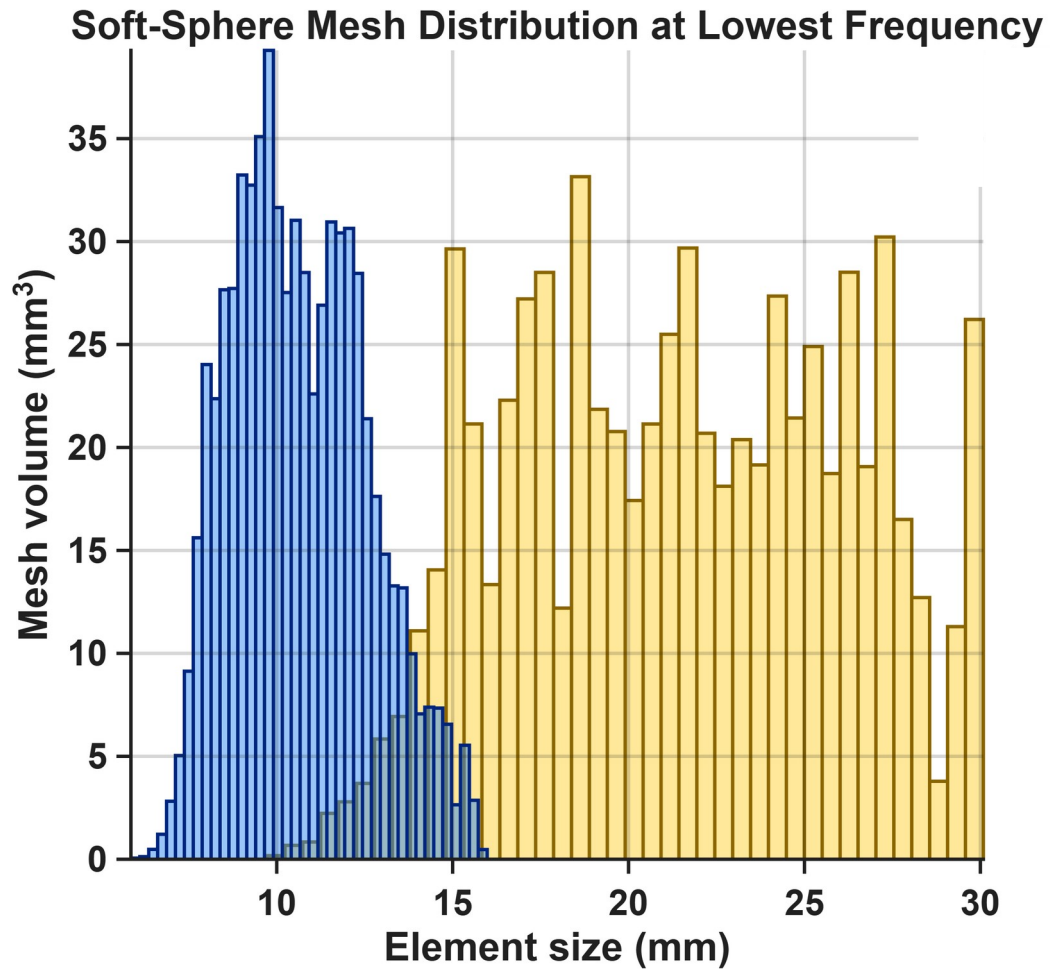


# **Mesh size comparisons**

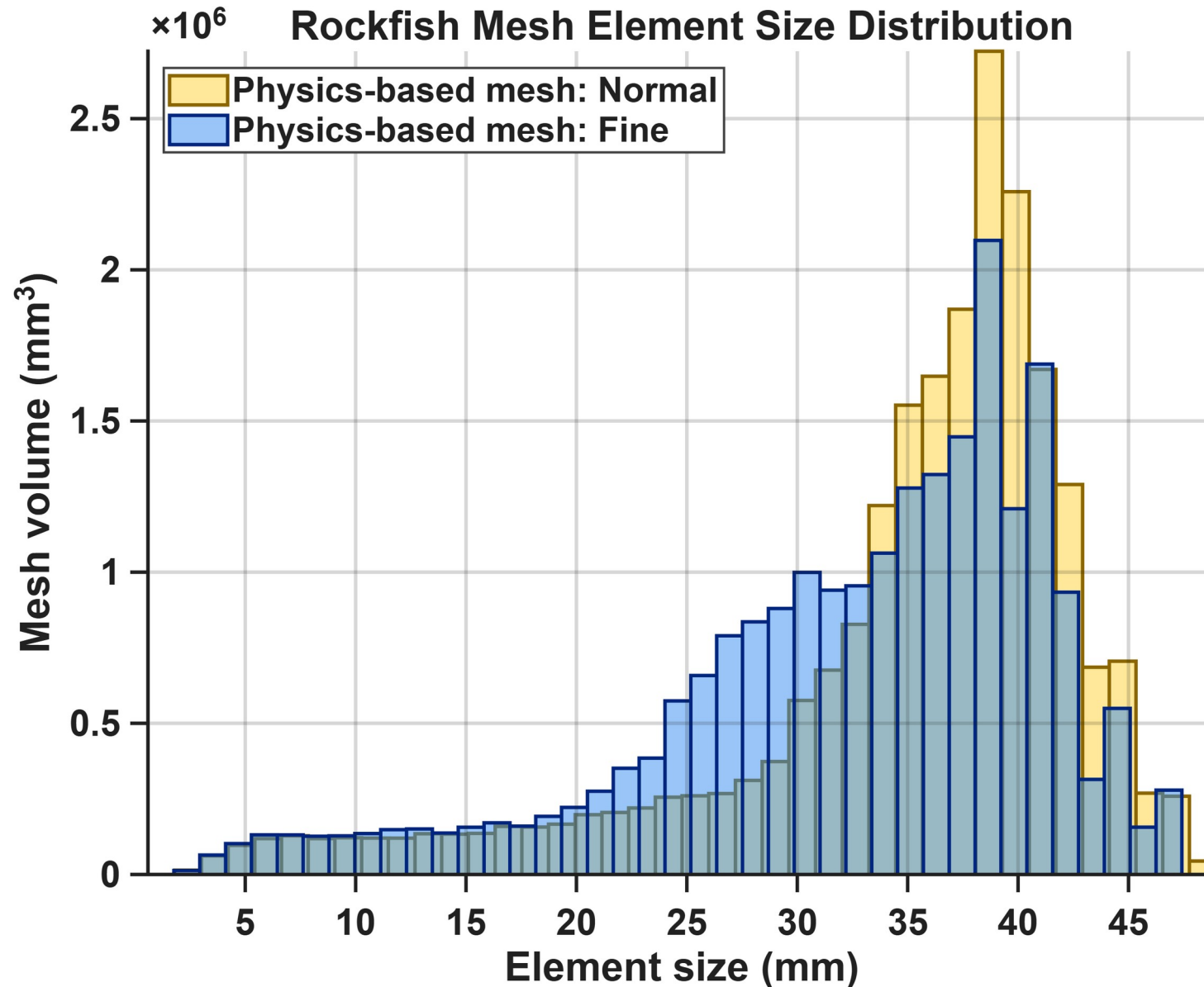
# Soft sphere



# Soft sphere



# Rockfish



## NOTE

These distributions are for the mesh used for the highest frequency in the sweep.

I have not computed user-based mesh statistics for this one yet, but still interesting to see comparisons between fine and normal